

What Is Value at Risk (VaR) and How to Calculate It?

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Value at Risk (VaR) has been called the "new science of risk management," and is a statistic that is used to predict the greatest possible losses over a specific time frame.

Commonly used by financial firms and commercial banks in investment analysis, VaR can determine the extent and probabilities of potential losses in portfolios. Risk managers use VaR to measure and control the level of risk exposure.

In Part 1, let's calculate VaR for the Nasdaq 100 index (QQQ) and establish that VaR answers a three-part question: "What is the worst loss that I can expect during a specified period with a certain confidence level?"

Key Takeaways

- Value at Risk (VaR) is a statistic that is used in risk management to predict the greatest possible losses over a specific time frame.
- VaR is determined by three variables: period, confidence level, and the size of the possible loss.
- There are three methods of calculating Value at Risk (VaR) including the historical method, the variance-covariance method, and the Monte Carlo simulation.

Elements of Value at Risk (VaR)

The traditional measure of risk is volatility, and an investor's main concern is the odds of losing money. The VaR statistic has three components: a period, a confidence level, and a loss amount, or loss percentage, and can address these concerns:

- What can I expect to lose in dollars with a 95% or 99% level of confidence next month?
- What is the maximum percentage I can expect to lose with 95% or 99% confidence over the next year?

The questions include a high level of confidence, a period, and an estimate of investment loss.

Methods of Calculating VaR

Let's evaluate the risk of a single index that trades like a stock, the Nasdaq 100 Index, which is traded through the Invesco QQQ Trust. The QQQ is an index of the largest non-financial stocks that trade on the Nasdaq exchange.¹

There are three methods of calculating Value at Risk (VaR) including the historical method, the variance-covariance method, and the Monte Carlo simulation.²

1. Historical Method

The historical method simply re-organizes actual historical returns, putting them in order from worst to best. It then assumes that history will repeat itself, from a risk perspective. Let's look at the Nasdaq 100 ETF, which trades under the symbol QQQ.³

Value at Risk

Value at Risk = $v_m (v_i / v(i - 1))$

M = the number of days from which historical data is taken

v_i = the number of variables on the day i .

In calculating each daily return, we produce a rich data set of more than 1,400 points. Let's put them in a histogram that compares the frequency of return "buckets."

At the highest bar, there were more than 250 days when the daily return was between 0% and 1%. At the far right, a tiny bar at 13% represents the one single day within five-plus years when the daily return for the QQQ was 12.4%.⁴

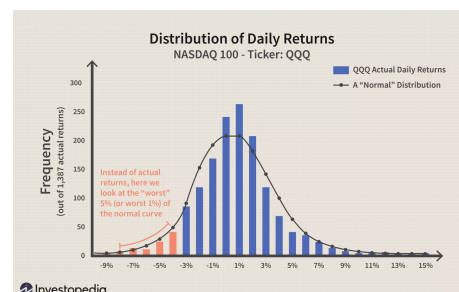


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2. The Variance-Covariance Method

This method assumes that stock returns are normally distributed and requires an estimate of only two factors, an expected return, and a standard deviation, allowing for a normal distribution curve. The normal curve is plotted against the same actual return data in the graph above.

The variance-covariance is similar to the historical method except it uses a familiar curve instead of actual data. The advantage of the normal curve is that it shows where the worst 5% and 1% lie on the curve. They are a function of desired confidence and the standard deviation.

Confidence	# of Standard Deviations (σ)
95% (high)	- 1.65 x σ
99% (really high)	- 2.33 x σ

The curve above is based on the actual daily standard deviation of the QQQ, which is 2.64%. The average daily return happened to be fairly close to zero, so it's safe to assume an average return of zero for illustrative purposes. Here are the results of using the actual standard deviation in the formulas above:

Confidence	# of σ	Calculation	Equals
95% (high)	- 1.65 x σ	- 1.65 x (2.64%) =	-4.36%
99% (really high)	- 2.33 x σ	- 2.33 x (2.64%) =	-6.15%

3. Monte Carlo Simulation

A Monte Carlo simulation refers to any method that randomly generates trials, but by itself does not tell us anything about the underlying methodology.

For most users, a Monte Carlo simulation amounts to a "black box" generator of random, probabilistic outcomes. This technique uses computational models to simulate projected returns over hundreds or thousands of possible iterations.

If 100 hypothetical trials of monthly returns for the QQQ were conducted, two of the worst outcomes may be between -15% and -20%, and three between -20% and 25%. That means the worst five outcomes were less than -15%.

The Monte Carlo simulation, therefore, leads to the following VaR-type conclusion: with 95% confidence, we do not expect to lose more than 15% during any given month.

What Is the Disadvantage of Using Value at Risk?

While VaR is useful for predicting the risks facing an investment, it can be misleading. One critique is that different methods give different results: you might get a gloomy forecast with the historical method while Monte Carlo Simulations are relatively optimistic. It can also be difficult to calculate the VaR for large portfolios: you can't simply calculate the VaR for each asset, since many of those assets will be correlated. Finally, any VaR calculation is only as good as the data and assumptions that go into it.⁵

What Are the Advantages of Using Value at Risk?

VaR is a single number that indicates the extent of risk in a given portfolio and is measured in either price or as a percentage, making understanding VaR easy. It can be applied to assets

such as bonds, shares, and currencies, and is used by banks and financial institutions to assess the profitability and risk of different investments, and allocate risk based on VaR.

What Does a High VaR Mean?

A high value for the confidence interval percentage means greater confidence in the likelihood of the projected outcome. Alternatively, a high value for the projected outcome is not ideal and statistically anticipates a higher dollar loss to occur.

The Bottom Line

Value at Risk (VAR) calculates the maximum loss expected on an investment over a given period and given a specified degree of confidence. We looked at three methods commonly used to calculate VAR. In Part 2 of this series, we show you how to compare different time horizons.